The Euclidean Algorithm:

Abstract

The Euclidean algorithm computes the greatest common divisor (GCD) of two integers efficiently by repeated division and remainders. Originating from Euclid's *Elements* around 300 BC, it underpins number theory, cryptography, and computational mathematics. This paper reviews its history, proves its correctness and termination, outlines implementations, and discusses key applications.[1]

Introduction

The Euclidean algorithm, often called Euclid's algorithm, finds the GCD of integers a and b without prime factorization. It relies on the property that $\gcd(a, b) = \gcd(b, a \bmod b)$, iterating until the remainder is zero. Named after Euclid, it remains foundational in modern computing despite its antiquity.[2][3][1]

Literature Review

Euclid described the algorithm in *Elements* Book VII (integers) and Book X (lengths), likely drawing from earlier sources like Pythagoreans or Theaetetus. Indian astronomers like Aryabhata (5th century) used it as the "pulverizer" for Diophantine equations, while Chinese texts like Qin Jiushao's 1247 work applied it to calendars. In the 19th century, Gauss and Dirichlet extended it to Gaussian integers and Euclidean domains; Lamé (1844) analyzed its complexity.[4][1][2]

Mathematical Theory / Proof

The algorithm generates remainders via $r_{k-2} = q_k r_{k-1} + r_k$ where $0 \leq r_k < |r_{k-1}|$, terminating when $r_n = 0$, so $\gcd(a, b) = r_{n-1}$ [1].

Termination Proof: Remainders form a strictly decreasing sequence of non-negative integers: $r_0 > r_1 > \cdots > r_{n-1} > r_n = 0$, which must end as no infinite descent exists in naturals.[5][1]

Correctness Proof: r_{n-1} divides all prior remainders by back-substitution, hence divides a and b. Any common divisor d of a and b divides all remainders, including r_{n-1}. Thus, r_{n-1} is the greatest such divisor.[6][1]

The extended version yields Bézout coefficients: $\gcd(a, b) = sa + tb$ via recursive back-substitution.[1]

Methodology

Implement as iteration: while $b \neq 0$, set $t = b$, $b = a \bmod b$, $a = t$; return a. For example, gcd(1071, 462):[1]

Step	Equation	Quotient (q)	Remainder (r)	
0	$1071 = q_0 \cdot 462 + r_0$	2	147	
1	$462 = q_1 \cdot 147 + r_1$	3	21	
2	$147 = q_2 \cdot 21 + r_2$	7	0	[1]

Complexity is $O(\log \min(a,b))$, worst-case on consecutive Fibonacci numbers.[7][4][2]

Applications

Reduces fractions, solves linear Diophantine equations ($ax + by = c$), computes modular inverses for RSA cryptography, and supports Chinese Remainder Theorem. In abstract algebra, it defines Euclidean domains like Gaussian integers; also used in continued fractions, polynomial GCDs, and error-correcting codes.[8][1]

Results & Discussion

The algorithm's logarithmic efficiency makes it ideal for large integers, outperforming naive methods. Lamé's theorem bounds steps by $\sim 1.44 \log_{10}(b)$ for $b < 10^d$. Modern variants like binary GCD optimize further, but the core remains unchanged; its role in proving unique prime factorization via Euclid's lemma highlights its theoretical depth.[9][2][1]

Conclusion

The Euclidean algorithm exemplifies timeless mathematical elegance, bridging ancient geometry and contemporary computing.

References

- Wikipedia: Euclidean algorithm. https://en.wikipedia.org/wiki/Euclidean_algorithm[1]
- Full content from https://en.wikipedia.org/wiki/Euclidean_algorithm[1]
- Full content from https://mathworld.wolfram.com/EuclideanAlgorithm.html[4]
- MathWorld: Euclidean Algorithm. https://mathworld.wolfram.com/EuclideanAlgorithm.html[2]
- ProofWiki: Euclidean Algorithm. https://proofwiki.org/wiki/Euclidean_Algorithm[5]
- Applications of the Euclidean algorithm. https://lirias.kuleuven.be/retrieve/5948[8]

- Euclidean Algorithm GCD. https://www.techchorus.net/posts/euclids-algorithm-gcd/[7]
- History of Euclidean Algorithm. https://www.academia.edu/98676921/The_History_of_Euclidean_Algorithm[3]
- Proof Euclidean Algorithm Works. https://www.cs.purdue.edu/homes/ssw/cs355/eucprf.pdf[6]
- Time Complexity Euclidean Algorithm. https://www.geeksforgeeks.org/dsa/time-complexity-of-euclidean-algorithm/[9]

⁂

1. https://en.wikipedia.org/wiki/Euclidean_algorithm
2. https://mathworld.wolfram.com/EuclideanAlgorithm.html
3. https://www.academia.edu/98676921/The_History_of_Euclidean_Algorithm
4. https://www.math.brown.edu/tbanchof/Beyond3d/chapter9/section01.html
5. https://proofwiki.org/wiki/Euclidean_Algorithm
6. https://www.cs.purdue.edu/homes/ssw/cs355/eucprf.pdf
7. https://www.techchorus.net/posts/euclids-algorithm-gcd/
8. https://lirias.kuleuven.be/retrieve/5948
9. https://www.geeksforgeeks.org/dsa/time-complexity-of-euclidean-algorithm/
10. https://www.math.uci.edu/~ndonalds/math161/euclid.pdf
11. https://arxiv.org/abs/physics/0406026
12. https://en.wikipedia.org/wiki/Euclidean_geometry
13. https://www.youtube.com/watch?v=GR964SKoLag
14. https://www.britannica.com/science/Euclidean-geometry
15. https://www.geeksforgeeks.org/dsa/euclidean-algorithms-basic-and-extended/
16. https://quicktakes.io/learn/mathematics/questions/can-you-explain-the-euclidean-method-and-its-impact-on-physics

17. https://lirias.kuleuven.be/retrieve/ca59bac5-78eb-4552-80e2-cfb3b5fa3bdc

18. https://www.sciencedirect.com/topics/mathematics/euclidean-algorithm

19. https://www.youtube.com/watch?v=H_2_nqKAZ5w

Manhattan Distance:

Abstract

Manhattan distance, or taxicab distance (L^1 metric), measures the sum of absolute coordinate differences between points, modeling grid-constrained paths. Named after Manhattan's street grid, it forms a non-Euclidean geometry with diamonds as "circles" and finds use in machine learning, pathfinding, and optimization. This paper explores its history, proves it as a norm, details computations, and reviews applications.[1][2]

Introduction

Manhattan distance between points [Equation] and [Equation] is [Equation] [2][3]. Unlike Euclidean distance, it ignores diagonals, suiting rectilinear movement like urban navigation. Popularized as taxicab geometry, it contrasts Euclidean spaces while satisfying metric axioms [1].

Literature Review

Roger Boscovich used the L^1 metric in 1757 regression analysis. Hermann Minkowski and Frigyes Riesz formalized it in 1910 amid non-Euclidean geometries and L^p spaces. Karl Menger coined "taxicab geometry" in 1952 for public education. Modern texts like *Taxicab Geometry* by Krause (1986) and applications in data science expanded its scope.[4][2]

Mathematical Theory / Proof

Metric Properties: Non-negativity: [Equation], zero iff [Equation]. Symmetry: [Equation]. Triangle inequality: [Equation] holds as [Equation] sums over i [5][2].

Norm Properties (L^1 Norm): [Equation] is positive definite, homogeneous ([Equation]), and subadditive [5].

In 2D, unit circle is a diamond with circumference 8 (π analog = 4); arc length [Equation] [2].

Methodology

Compute iteratively: for points (1,2) and (4,6), [Equation] [4]. In code or grids, sum absolutes; efficient O(n) [6].

Example distances from (0,0):

Point	Euclidean	Manhattan	
(1,0)	1	1	
(1,1)	√2 ≈1.41	2	
(2,3)	√13≈3.61	5	[2]

Applications

Serves as heuristic in A* pathfinding for grids/games, distance in k-NN clustering (robust to outliers), LASSO regression regularization, and sparse signal recovery in compressed sensing. Used in RNA analysis, delivery routing, and high-dimensional data.[7][8][2][4]

Results & Discussion

Manhattan outperforms Euclidean in sparse/high-dimensional data, reducing outlier sensitivity; e.g., in k-means, yields balanced clusters. Orientation-dependent (rotations alter distances), limiting generality but ideal for axis-aligned problems. Computational cost linear, scalable to big data.[6][4]

Conclusion

Manhattan distance provides a practical metric for grid worlds and robust statistics, complementing Euclidean norms.

References

- Wikipedia: Taxicab geometry. https://en.wikipedia.org/wiki/Taxicab_geometry[1]
- Full content from https://en.wikipedia.org/wiki/Taxicab_geometry[2]
- Full content from https://en.wikipedia.org/wiki/Manhattan_distance[3]
- YouTube: Prove Manhattan norm. https://www.youtube.com/watch?v=mIS7aMECB20[5]
- GeeksforGeeks: Manhattan Distance. https://www.geeksforgeeks.org/data-science/manhattan-distance/[4]
- Design Gurus: Manhattan in ML. https://www.designgurus.io/answers/detail/what-is-manhattan-distance-in-machine-learning[7]

- Activeloop: Manhattan Distance. https://www.activeloop.ai/resources/glossary/manhattan-distance/[8]
- DataCamp: Manhattan Distance. https://www.datacamp.com/tutorial/manhattan-distance[6]

⁂

1. https://en.wikipedia.org/wiki/Taxicab_geometry
2. https://www.math.uci.edu/~ndonalds/math161/euclid.pdf
3. https://mathworld.wolfram.com/EuclideanAlgorithm.html
4. https://www.geeksforgeeks.org/data-science/manhattan-distance/
5. https://www.youtube.com/watch?v=mIS7aMECB20
6. https://www.datacamp.com/tutorial/manhattan-distance
7. https://www.designgurus.io/answers/detail/what-is-manhattan-distance-in-machine-learning
8. https://www.activeloop.ai/resources/glossary/manhattan-distance/
9. https://catalog.manhattan.edu/undergraduate/artsandscience/maph/
10. https://en.wikipedia.org/?title=Manhattan_metric
11. https://beckassets.blob.core.windows.net/product/readingsample/836423/9783642147081_excerpt_001.pdf
12. https://www.nyc.gov/site/law/index.page
13. https://edgeinducedcohesion.blog/2025/12/21/white-paper-proofs-and-importance-of-manhattan-distance/
14. https://www.reddit.com/r/math/comments/1ieswog/why_doesnt_the_manhattan_distance_approximate_the/
15. https://reu.physics.ucsb.edu/sites/default/files/sitefiles/REUPapersTalks/2021-REUPapersTalks/Brian-Bauer-UCSB-REU-paper.pdf
16. https://uplatz.com/blog/manhattan-distance-formula-grid-based-metric-for-similarity-in-high-dimensions/
17. https://sites.science.oregonstate.edu/~tevian/physics/taxicab/taxicab.pdf
18. https://www.mathcalcs.org/manhattan-norm
19. https://digitalcommons.unl.edu/cgi/viewcontent.cgi?article=1047&context=mathmidexppap
20. https://www.sciencedirect.com/topics/mathematics/manhattan-distance
21. https://www.rosenfeldlaw.com

Minkowski's Theorem:

Abstract

Minkowski's theorem states that any centrally symmetric convex set in [Equation] with volume greater than [Equation] contains a non-zero integer lattice point. Proved by Hermann Minkowski in 1896, it foundationalizes the geometry of numbers, enabling proofs of finiteness in class groups and

Diophantine approximations. This paper details its history, proof, methodology, and applications in number theory.[1][2]

Introduction

Minkowski's theorem links continuous geometry to discrete lattices: For symmetric convex [Equation] with [Equation], [Equation] contains nonzero [Equation] (here [Equation], [Equation]). It quantifies how large convex sets must intersect lattices, birthing geometry of numbers.[2][1]

Literature Review

Hermann Minkowski introduced it in 1896, building on earlier lattice studies by Hermite and Korkine/Zolotarev. Proofs evolved via pigeonhole principle and Blichfeldt's 1914 strengthening for non-symmetric sets. Texts like Cassels' *Geometry of Numbers* (1959) and Matoušek's *Lectures on Discrete Geometry* (2002) standardize it; applications span Dirichlet's unit theorem to Minkowski-Hasse principle.[3][4][5][1]

Mathematical Theory / Proof

Theorem: Let [Equation] be convex, [Equation]-symmetric ([Equation]), closed, with [Equation]. Then [Equation].[1]

Proof Sketch (Pigeonhole): Consider fundamental parallelepiped [Equation], [Equation]. Map [Equation] by [Equation] (mod 2). [Equation] convex, [Equation], so by pigeonhole, [Equation], [Equation], [Equation] (convexity), [Equation], [Equation]. Then [Equation] (symmetry), lattice point.[6][1]

Sharpness: Open cube [Equation] has vol [Equation] for n>0, contains only origin.[1]

Methodology

To apply: Verify convexity/symmetry of [Equation], compute vol via integration or determinants. For lattices, scale by [Equation]. Example n=2: Disk [Equation], area 2 = 2^2 /2 <4? Wait, diamond area 2<4, no point; enlarge to area>4 guarantees [1].

n	Critical Vol	Example Set
1	2	(-1,1)
2	4	Diamond area 4

3	8	Octahedron	[1]

Applications

Proves Lagrange's four-squares theorem (sum 4 squares <4p implies =p mod p for prime p=4k+1). Yields finiteness of ideal class groups in algebraic number theory, Dirichlet's unit theorem, and bounds on linear forms in logarithms. Extends to successive minima and reduction theory.[7][8][5][9]

Results & Discussion

Theorem bounds shortest lattice vectors: [Equation]. Sharp in low dimensions but loose asymptotically; variants like second theorem refine products of minima. Impacts cryptography (lattice-based crypto hardness) and optimization.[10][6][1]

Conclusion

Minkowski's theorem elegantly fuses geometry and arithmetic, powering key results in number theory.

References

- Wikipedia: Minkowski's theorem. https://en.wikipedia.org/wiki/Minkowski's_theorem[1]
- Minkowski's Theorem Notes. https://www.math.cmu.edu/~ttkocz/teaching/1819/read-sem-notes.pdf[3]
- Encyclopedia of Math: Minkowski theorem. https://encyclopediaofmath.org/wiki/Minkowski_theorem[2]
- Three proofs Minkowski inequality. https://www.cambridge.org/core/services/aop-cambridge-core/content/view/F132AFA5607313F0969714D93697F20A/S1446788700028482a.pdf/[10]
- Geometry of Numbers Notes. http://www.rvirk.com/notes/student/minkowski.pdf[8]
- Archive of Formal Proofs: Minkowski's Theorem. https://www.isa-afp.org/entries/Minkowskis_Theorem.html[6]
- Exploration Minkowski Theory. http://math.uchicago.edu/~may/REU2012/REUPapers/Comeaux.pdf[5]
- Geometry of Numbers Chapter. https://pub.math.leidenuniv.nl/~evertsejh/dio17-2.pdf[9]

⁂

1. https://en.wikipedia.org/wiki/Minkowski's_theorem

2. https://encyclopediaofmath.org/wiki/Minkowski_theorem
3. https://www.math.cmu.edu/~ttkocz/teaching/1819/read-sem-notes.pdf
4. https://www.math.cmu.edu/~ttkocz/teaching/1819/read-sem-notes
5. http://math.uchicago.edu/~may/REU2012/REUPapers/Comeaux.pdf
6. https://www.isa-afp.org/entries/Minkowskis_Theorem.html
7. https://www.epfl.ch/labs/disopt/wp-content/uploads/2018/09/minkowski.pdf
8. http://www.rvirk.com/notes/student/minkowski.pdf
9. https://pub.math.leidenuniv.nl/~evertsejh/dio17-2.pdf
10. https://www.cambridge.org/core/services/aop-cambridge-core/content/view/F132AFA5607313F0969714D93697F20A/S1446788700028482a.pdf/three-proofs-of-minkowskis-second-inequality-in-the-geometry-of-numbers.pdf
11. https://faculty.gardner.wwu.edu/gorizia12.pdf
12. https://www.math.union.edu/~hatleyj/Capstone.pdf
13. https://en.wikipedia.org/wiki/Minkowski_space
14. https://arxiv.org/abs/2010.07187
15. https://www.wstein.org/edu/2010/414/projects/itolkov.pdf
16. https://www.cambridge.org/core/books/geometry-of-numbers/minkowskis-fundamental-theorem/5EB5835C1213EF7502E53F1A5EE05BDF
17. https://en.wikipedia.org/wiki/Brunn–Minkowski_theorem

what the difference is between Euclidean, Manhattan, Minkowski

Euclidean, Manhattan, and Minkowski refer to distance metrics (not "laws" per se), with Euclidean and Manhattan as specific cases and Minkowski as the general form. These metrics quantify "distance" between points in n-dimensional space, differing in computation and geometry [page:1 from prior].[1]

Core Formulas

All compare points [Equation] and [Equation].

- **Euclidean (L_2)**: [Equation] – straight-line "as the crow flies."
- **Manhattan (L_1, taxicab)**: [Equation] – grid-path sum, no diagonals.
- **Minkowski (L_p)**: [Equation] – generalizes both [1][2].

Key Differences

Aspect	Euclidean (p=2)	Manhattan (p=1)	Minkowski (general p)

Geometry	Circles as round disks.	"Circles" as diamonds.	p<2: diamond-like; p>2: square-like [1].
Path Model	Shortest straight path.	Axis-aligned (streets/urban).	Tunable; approximates others.
Sensitivity	Quadratic: outliers amplify.	Linear: robust to outliers.	p controls: low p milder [1].
Example (pts (0,0),(3,4))	5 (Pythagoras).	7 (3+4).	p=1:7; p=2:5; p=∞:4 (max diff).

When to Use

- Euclidean: Natural spaces (physics, images).
- Manhattan: Grids, sparse data, ML (k-NN robust).[1]
- Minkowski: Flexible parameter p for experiments.[3]

Euclidean/Manhattan are Minkowski specials (p=2/1); prior papers covered Euclidean algorithm (GCD), taxicab geometry, Minkowski's lattice theorem—here, conversation context aligns on distances.[1]

⁂

1. https://www.kdnuggets.com/2023/03/distance-metrics-euclidean-manhattan-minkowski-oh.html
2. https://en.wikipedia.org/wiki/Minkowski_distance
3. https://www.youtube.com/watch?v=0kr0xBZ4AOk
4. https://www.youtube.com/watch?v=BjKrvzlu018
5. https://research.ijcaonline.org/volume67/number10/pxc3886785.pdf
6. https://ijsrset.com/paper/6543.pdf
7. https://www.kaggle.com/code/nishtha711/visualizing-distance-metrics-in-ml
8. https://ieeexplore.ieee.org/document/9392053/
9. https://www.youtube.com/watch?v=gwtTM9M6ugU
10. https://www.youtube.com/watch?v=Gc7otqkPq2g